The book walkthrough: one command, then Excel

What the firm asked for, 14 Oct 2026: *"i want you to make it work so i can generate this workbook at work with the script, it should be debugged and walked through and tied out."*

This is the walk: the workbook built the way it is built at work, then used in Excel 16 the way a person uses it (clicking the buttons, typing in the yellow row, saving, closing and opening it again), and every answer read back from Excel and set beside an answer written down beforehand from the source files.

1. Build

From `threshold-engine/`:

```
python -m threshold_engine peers pull --cache books/ind_cache --first 2006Q1 --last 2026Q2 --products "Cre
dit card" Mortgage "Home equity" "Other consumer" Auto --delinquency yes --mergers 2005-01-01
python -m threshold_engine book books/consumer-public.json consumer-credit.xlsx
```

The walkthrough ran the second command on a copy of the book whose only change is where the cache is (`prepare.py`): the cache pulled on 14 Oct 2026 was used rather than pulled again. Expected: `Wrote consumer-credit.xlsx: 105 series, 82 quarters; open it on Summary, decide on Decide.` It took 8 min 48 s on the last run (12 min 42 s on the first), almost all of it the peers build reading the cache.

2. The steps

The steps are cumulative, as a session is: what is typed in one step is still there in the next.

One set of lines per product and measure (the firm, 15 Oct 2026: *"we don't need to set thresholds on all 3, we need to use all 3 to set appropriate thresholds"*): Decide has 35 rows, and each row's **Lines set from** picks the source whose history sets its lines.

Step	What you do	What it shows
B1	Open the workbook	It opens on Summary; the tabs in order (Assess, Chart and Peers workings hidden); two slicers, one button each; Decide shows the first product and measure, no source picked, nothing decided; no links, no errors
B2	Read Summary and Lines by source	All 105 series' latest value, Pick a source where each rating goes, and every series' own lines equal the reference
B3	On Decide, click Credit card and Nonaccrual, and pick KeyBank under Lines set from	One row; the lines are KeyBank's own; every source rated on them, beside what its own history would give; Thresholds holds the same
B4	Type 3.5 under How rare High is (z)	High's line moves for every source of the row; Thresholds and Summary follow
B5	Leave out 2012Q3 to 2013Q4, the period Decide suggests, with a reason	KeyBank's lines move as the reference says, the industry's own lines do not; Assess lists the period under KeyBank's series, whole
B6, B6b	Clear the reason; type it back	Refused, in words, on Decide and Thresholds; then the lines as in B5
B7	Accept these lines, with a note	Decided: 1 of 35; the decision on Thresholds, and on Lines by source beside KeyBank only
B8	Click Auto and Net charge-offs, and pick Industry (every bank)	KeyBank's life-to-date history rated on the lines the industry's longer history gives

Step	What you do	What it shows
B9	Click Mortgage and Net charge-offs, and pick Industry (every bank)	Refused, with the number that would fix it
B10	Save, close, open again	Every pick, decision and setting still there, and the lines with them

A step passes when every cell it reads says what `steps.json` says: a number within 1e-9, text exactly.

3. How it was run

```
python docs/walkthrough-book/prepare.py PATH-TO-CACHE
python -m threshold_engine book docs/walkthrough-book/out/book.json docs/walkthrough-book/out/consumer-cre
dit.xlsx
python docs/walkthrough-book/expected.py docs/walkthrough-book/out/book.json
copy docs\walkthrough-book\out\consumer-credit.xlsx docs\walkthrough-book\out\walk.xlsx
powershell -NoProfile -ExecutionPolicy Bypass -File docs\walkthrough-book\walk.ps1
python docs/walkthrough-book/compare.py
```

`expected.py` writes the answers from the series CSVs and `scale.py`, never from the workbook. `walk.ps1` does each step in Excel through Excel's own automation and reads each answer back. `compare.py` writes `RESULT.md`.

And, separately, that the file opens without Excel repairing it:

```
powershell -NoProfile -ExecutionPolicy Bypass -File tools\check_slicers_in_excel.ps1 -Path docs\walkthroug
h-book\out\consumer-credit.xlsx -Sheet Decide -Table Decisions -Shapes 3 -Expect "Slicer_Product=Credit ca
rd|Slicer_Measure=Net charge-offs" -SwitchCache Slicer_Measure -SwitchTo "Gross charge-offs"
```

"No errors" in B1 means none a reader can see: on the visible tabs, outside hidden rows and columns. The rest are counted by kind and are all #N/A, the gap a chart leaves where a quarter has no value (on the run of 15 Oct 2026: Assess 169, Chart 8,470, Decide 249, Peers workings 1,556).

4. What it does not prove

- The peer tabs are counted for errors and links (B1) but their numbers are not checked against a reference here; `tests/test_peers*.py` hold them.
- One series is tied to the filing (`../tie-out/keybank-card-nonaccrual-2026-10-14/`); the other 104 rest on the same code and the reference, not on a filing each.
- Three of the 35 rows have a source picked in the walk (Credit card nonaccrual, Auto and Mortgage net charge-offs); the other 32 are read blank (Pick a source) on Summary, and every source's own lines are read for all 105 on Lines by source.
- Thresholds' latest value is written by the build, not a live formula on Data: pasting a corrected quarter into Data moves the lines but not the latest value (found by the tie-out).

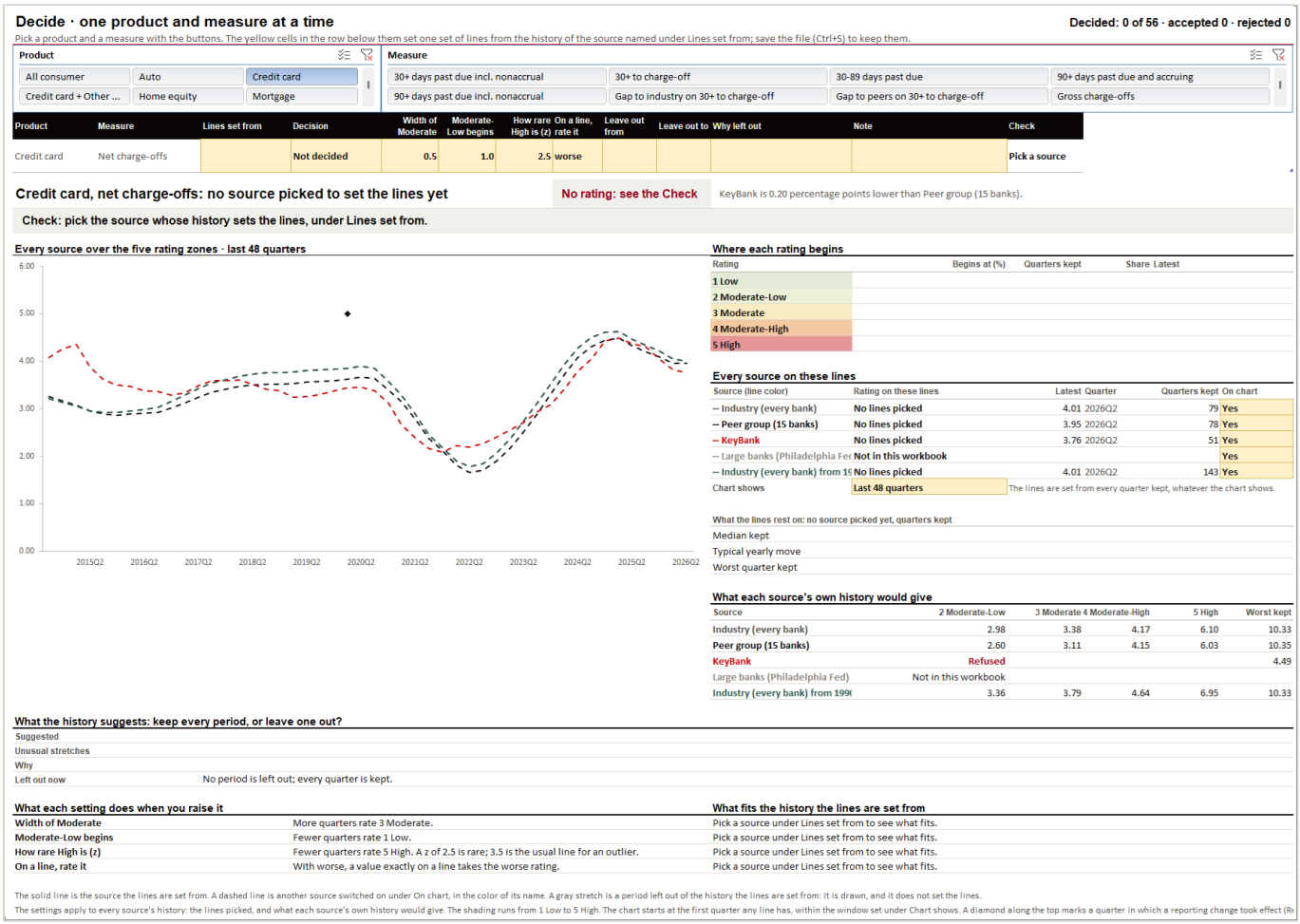
5. The result

2258 of 2258 reads as expected.

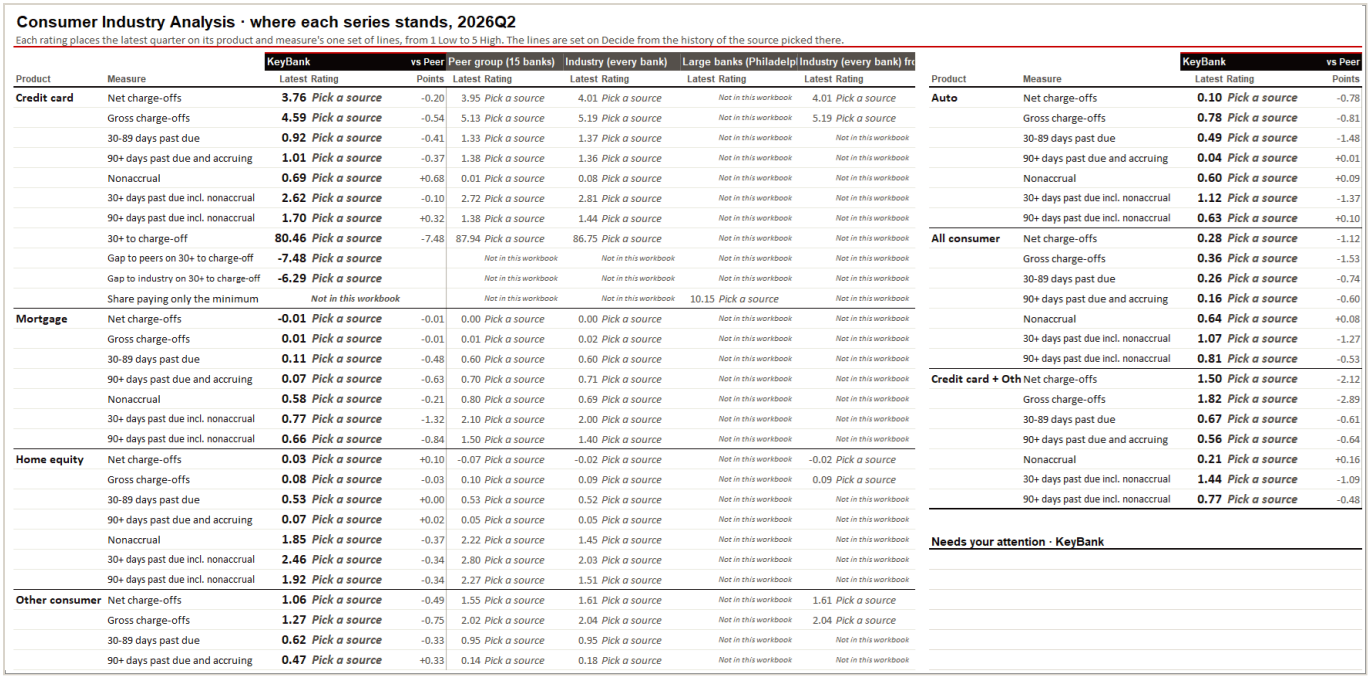
Step	What you do	Reads as expected
B1	Open the workbook as built	10 of 10
B2	Read Summary, and every source's own lines on Lines by source	1971 of 1971

B3	On Decide, click Credit card and Nonaccrual, and pick KeyBank under Lines set from	56 of 56
B4	Type 3.5 under How rare High is (z), in the yellow row	52 of 52
B5	Leave out 2012Q3 to 2013Q4, the period Decide suggests, with a reason	38 of 38
B6	Clear the reason	3 of 3
B6b	Type the reason back	23 of 23
B7	Pick Accept these lines and write a note	4 of 4
B8	Click Auto and Net charge-offs, and pick Industry (every bank) under Lines set from	28 of 28
B9	Click Mortgage and Net charge-offs, and pick Industry (every bank)	22 of 22
B10	Save (Ctrl+S), close Excel, open the file again	30 of 30
B11	On Peers, set Banks in the pack to Regionals only	6 of 6
B12	Set Banks in the pack back to Every peer, and JPMorgan's cell to No	6 of 6
B13	Set JPMorgan's cell back to Pack	9 of 9

6. What Excel showed



B1 · Open the workbook as built (Decide tab, photographed by Excel during the walk)



Consumer Industry Analysis · where each series stands, 2026Q2

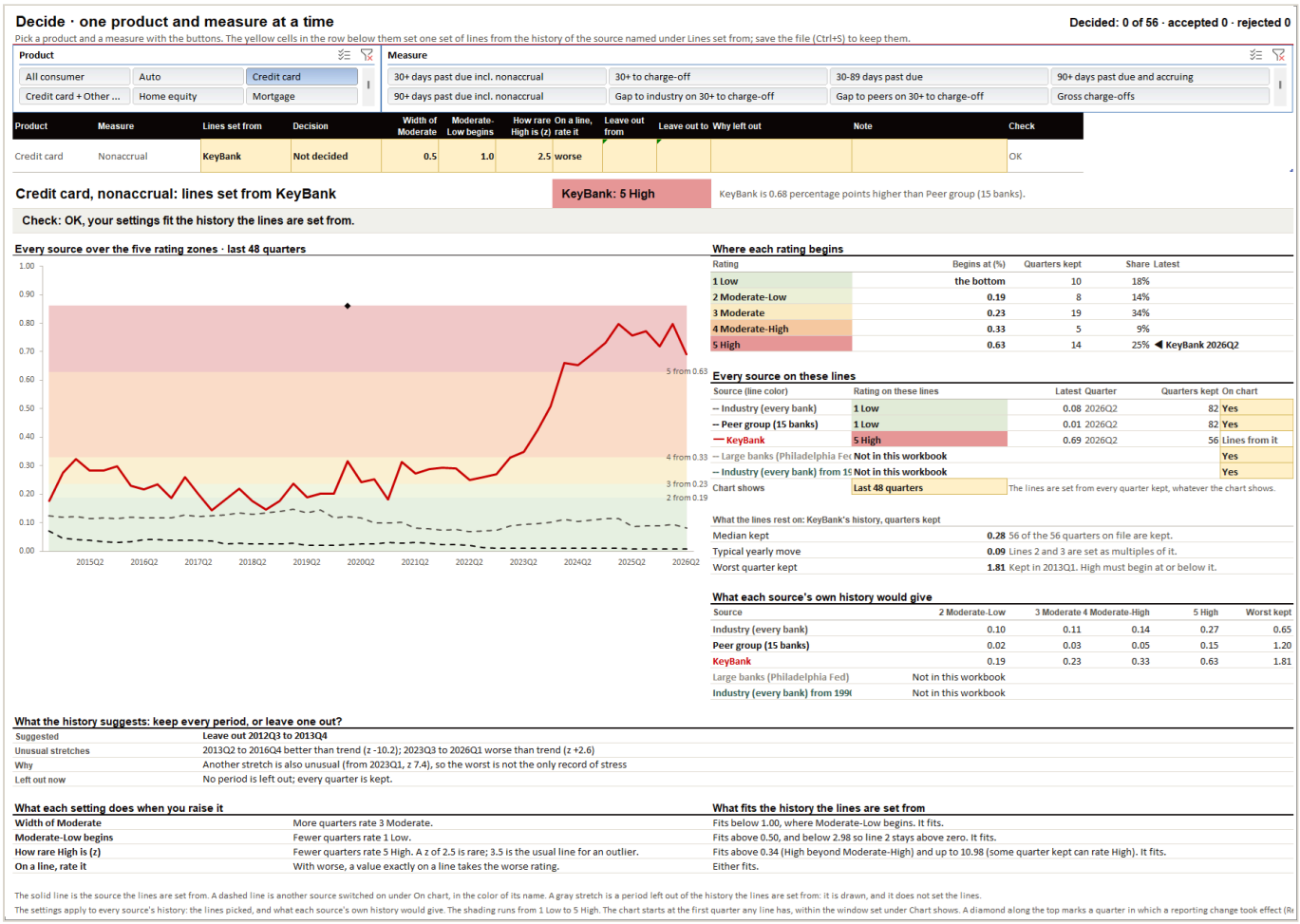
Each rating places the latest quarter on its product and measure's one set of lines, from 1 Low to 5 High. The lines are set on Decide from the history of the source picked there.

Product	Measure	KeyBank Latest	KeyBank Rating	vs Peer Points	Peer group (15 banks) Latest	Peer group (15 banks) Rating	Industry (every bank) Latest	Industry (every bank) Rating	Large banks (Philadelp Latest	Large banks (Philadelp Rating	Industry (every bank) fr Latest	Industry (every bank) fr Rating
Credit card	Net charge-offs	**3.76**	*Pick a source*	-0.20	3.95	*Pick a source*	4.01	*Pick a source*	*Not in this workbook*		4.01	*Pick a source*
	Gross charge-offs	**4.59**	*Pick a source*	-0.54	5.13	*Pick a source*	5.19	*Pick a source*	*Not in this workbook*		5.19	*Pick a source*
	30-89 days past due	**0.92**	*Pick a source*	-0.41	1.33	*Pick a source*	1.37	*Pick a source*	*Not in this workbook*		*Not in this workbook*	
	90+ days past due and accruing	**1.01**	*Pick a source*	-0.37	1.38	*Pick a source*	1.36	*Pick a source*	*Not in this workbook*		*Not in this workbook*	
	Nonaccrual	**0.69**	*Pick a source*	+0.68	0.01	*Pick a source*	0.08	*Pick a source*	*Not in this workbook*		*Not in this workbook*	
	30+ days past due incl. nonaccrual	**2.62**	*Pick a source*	-0.10	2.72	*Pick a source*	2.81	*Pick a source*	*Not in this workbook*		*Not in this workbook*	
	90+ days past due incl. nonaccrual	**1.70**	*Pick a source*	+0.32	1.38	*Pick a source*	1.44	*Pick a source*	*Not in this workbook*		*Not in this workbook*	
	30+ to charge-off	**80.46**	*Pick a source*	-7.48	87.94	*Pick a source*	86.75	*Pick a source*	*Not in this workbook*		*Not in this workbook*	
	Gap to peers on 30+ to charge-off	**-7.48**	*Pick a source*		*Not in this workbook*		*Not in this workbook*		*Not in this workbook*		*Not in this workbook*	
	Gap to industry on 30+ to charge-off	**-6.29**	*Pick a source*		*Not in this workbook*		*Not in this workbook*		*Not in this workbook*		*Not in this workbook*	
	Share paying only the minimum		*Not in this workbook*		*Not in this workbook*		*Not in this workbook*		10.15	*Pick a source*	*Not in this workbook*	
Mortgage	Net charge-offs	**-0.01**	*Pick a source*	-0.01	0.00	*Pick a source*	0.00	*Pick a source*	*Not in this workbook*		*Not in this workbook*	
	Gross charge-offs	**0.01**	*Pick a source*	-0.01	0.01	*Pick a source*	0.02	*Pick a source*	*Not in this workbook*		*Not in this workbook*	
	30-89 days past due	**0.11**	*Pick a source*	-0.48	0.60	*Pick a source*	0.60	*Pick a source*	*Not in this workbook*		*Not in this workbook*	
	90+ days past due and accruing	**0.07**	*Pick a source*	-0.63	0.70	*Pick a source*	0.71	*Pick a source*	*Not in this workbook*		*Not in this workbook*	
	Nonaccrual	**0.58**	*Pick a source*	-0.21	0.80	*Pick a source*	0.69	*Pick a source*	*Not in this workbook*		*Not in this workbook*	
	30+ days past due incl. nonaccrual	**0.77**	*Pick a source*	-1.32	2.10	*Pick a source*	2.00	*Pick a source*	*Not in this workbook*		*Not in this workbook*	
	90+ days past due incl. nonaccrual	**0.66**	*Pick a source*	-0.84	1.50	*Pick a source*	1.40	*Pick a source*	*Not in this workbook*		*Not in this workbook*	
Home equity	Net charge-offs	**0.03**	*Pick a source*	+0.10	-0.07	*Pick a source*	-0.02	*Pick a source*	*Not in this workbook*		-0.02	*Pick a source*
	Gross charge-offs	**0.08**	*Pick a source*	-0.03	0.10	*Pick a source*	0.09	*Pick a source*	*Not in this workbook*		0.09	*Pick a source*
	30-89 days past due	**0.53**	*Pick a source*	+0.00	0.53	*Pick a source*	0.52	*Pick a source*	*Not in this workbook*		*Not in this workbook*	
	90+ days past due and accruing	**0.07**	*Pick a source*	+0.02	0.05	*Pick a source*	0.05	*Pick a source*	*Not in this workbook*		*Not in this workbook*	
	Nonaccrual	**1.85**	*Pick a source*	-0.37	2.22	*Pick a source*	1.45	*Pick a source*	*Not in this workbook*		*Not in this workbook*	
	30+ days past due incl. nonaccrual	**2.46**	*Pick a source*	-0.34	2.80	*Pick a source*	2.03	*Pick a source*	*Not in this workbook*		*Not in this workbook*	
	90+ days past due incl. nonaccrual	**1.92**	*Pick a source*	-0.34	2.27	*Pick a source*	1.51	*Pick a source*	*Not in this workbook*		*Not in this workbook*	
Other consumer	Net charge-offs	**1.06**	*Pick a source*	-0.49	1.55	*Pick a source*	1.61	*Pick a source*	*Not in this workbook*		1.61	*Pick a source*
	Gross charge-offs	**1.27**	*Pick a source*	-0.75	2.02	*Pick a source*	2.04	*Pick a source*	*Not in this workbook*		2.04	*Pick a source*
	30-89 days past due	**0.62**	*Pick a source*	-0.33	0.95	*Pick a source*	0.95	*Pick a source*	*Not in this workbook*		*Not in this workbook*	
	90+ days past due and accruing	**0.47**	*Pick a source*	+0.33	0.14	*Pick a source*	0.18	*Pick a source*	*Not in this workbook*		*Not in this workbook*	

Product	Measure	KeyBank Latest	KeyBank Rating	vs Peer Points
Auto	Net charge-offs	**0.10**	*Pick a source*	-0.78
	Gross charge-offs	**0.78**	*Pick a source*	-0.81
	30-89 days past due	**0.49**	*Pick a source*	-1.48
	90+ days past due and accruing	**0.04**	*Pick a source*	+0.01
	Nonaccrual	**0.60**	*Pick a source*	+0.09
	30+ days past due incl. nonaccrual	**1.12**	*Pick a source*	-1.37
	90+ days past due incl. nonaccrual	**0.63**	*Pick a source*	+0.10
All consumer	Net charge-offs	**0.28**	*Pick a source*	-1.12
	Gross charge-offs	**0.36**	*Pick a source*	-1.53
	30-89 days past due	**0.26**	*Pick a source*	-0.74
	90+ days past due and accruing	**0.16**	*Pick a source*	-0.60
	Nonaccrual	**0.64**	*Pick a source*	+0.08
	30+ days past due incl. nonaccrual	**1.07**	*Pick a source*	-1.27
	90+ days past due incl. nonaccrual	**0.81**	*Pick a source*	-0.53
Credit card + Oth	Net charge-offs	**1.50**	*Pick a source*	-2.12
	Gross charge-offs	**1.82**	*Pick a source*	-2.89
	30-89 days past due	**0.67**	*Pick a source*	-0.61
	90+ days past due and accruing	**0.56**	*Pick a source*	-0.64
	Nonaccrual	**0.21**	*Pick a source*	+0.16
	30+ days past due incl. nonaccrual	**1.44**	*Pick a source*	-1.09
	90+ days past due incl. nonaccrual	**0.77**	*Pick a source*	-0.48

Needs your attention · KeyBank

B1 · Open the workbook as built (Summary tab, photographed by Excel during the walk)



Decide · one product and measure at a time

Decided: 0 of 56 · accepted 0 · rejected 0

Pick a product and a measure with the buttons. The yellow cells in the row below them set one set of lines from the history of the source named under Lines set from; save the file (Ctrl+S) to keep them.

Product: All consumer | Auto | Credit card | Credit card + Other ... | Home equity | Mortgage

Measure: 30+ days past due incl. nonaccrual | 30+ to charge-off | 30-89 days past due | 90+ days past due and accruing | 90+ days past due incl. nonaccrual | Gap to industry on 30+ to charge-off | Gap to peers on 30+ to charge-off | Gross charge-offs

Product	Measure	Lines set from	Decision	Width of Moderate	Moderate-Low begins	How rare High is (z)	On a line, rate it	Leave out from	Leave out to	Why left out	Note	Check
Credit card	Nonaccrual	KeyBank	Not decided	0.5	1.0	2.5	worse					OK

Credit card, nonaccrual: lines set from KeyBank

KeyBank: 5 High — KeyBank is 0.68 percentage points higher than Peer group (15 banks).

Check: OK, your settings fit the history the lines are set from.

Every source over the five rating zones · last 48 quarters

Where each rating begins

Rating	Begins at (%)	Quarters kept	Share	Latest
1 Low	the bottom	10	18%	
2 Moderate-Low	0.19	8	14%	
3 Moderate	0.23	19	34%	
4 Moderate-High	0.33	5	9%	
5 High	0.63	14	25%	◀ KeyBank 2026Q2

Every source on these lines

Source (line color)	Rating on these lines	Latest	Quarter	Quarters kept	On chart
-- Industry (every bank)	1 Low	0.08	2026Q2	82	Yes
-- Peer group (15 banks)	1 Low	0.01	2026Q2	82	Yes
— KeyBank	5 High	0.69	2026Q2	56	Lines from it
-- Large banks (Philadelphia Fed	Not in this workbook				Yes
-- Industry (every bank) from 19	Not in this workbook				Yes
Chart shows	Last 48 quarters	The lines are set from every quarter kept, whatever the chart shows.			

What the lines rest on: KeyBank's history, quarters kept

Median kept	**0.28**	56 of the 56 quarters on file are kept.
Typical yearly move	**0.09**	Lines 2 and 3 are set as multiples of it.
Worst quarter kept	**1.81**	Kept in 2013Q1. High must begin at or below it.

What each source's own history would give

Source	2 Moderate-Low	3 Moderate 4 Moderate-High	5 High	Worst kept
Industry (every bank)	0.10	0.11 0.14	0.27	0.65
Peer group (15 banks)	0.02	0.03 0.05	0.15	1.20
KeyBank	0.19	0.23 0.33	0.63	1.81
Large banks (Philadelphia Fed)	Not in this workbook			
Industry (every bank) from 199	Not in this workbook			

What the history suggests: keep every period, or leave one out?

Suggested	**Leave out 2012Q3 to 2013Q4**
Unusual stretches	2013Q2 to 2016Q4 better than trend (z -10.2); 2023Q3 to 2026Q1 worse than trend (z +2.6)
Why	Another stretch is also unusual (from 2023Q1, z 7.4), so the worst is not the only record of stress
Left out now	No period is left out; every quarter is kept.

What each setting does when you raise it

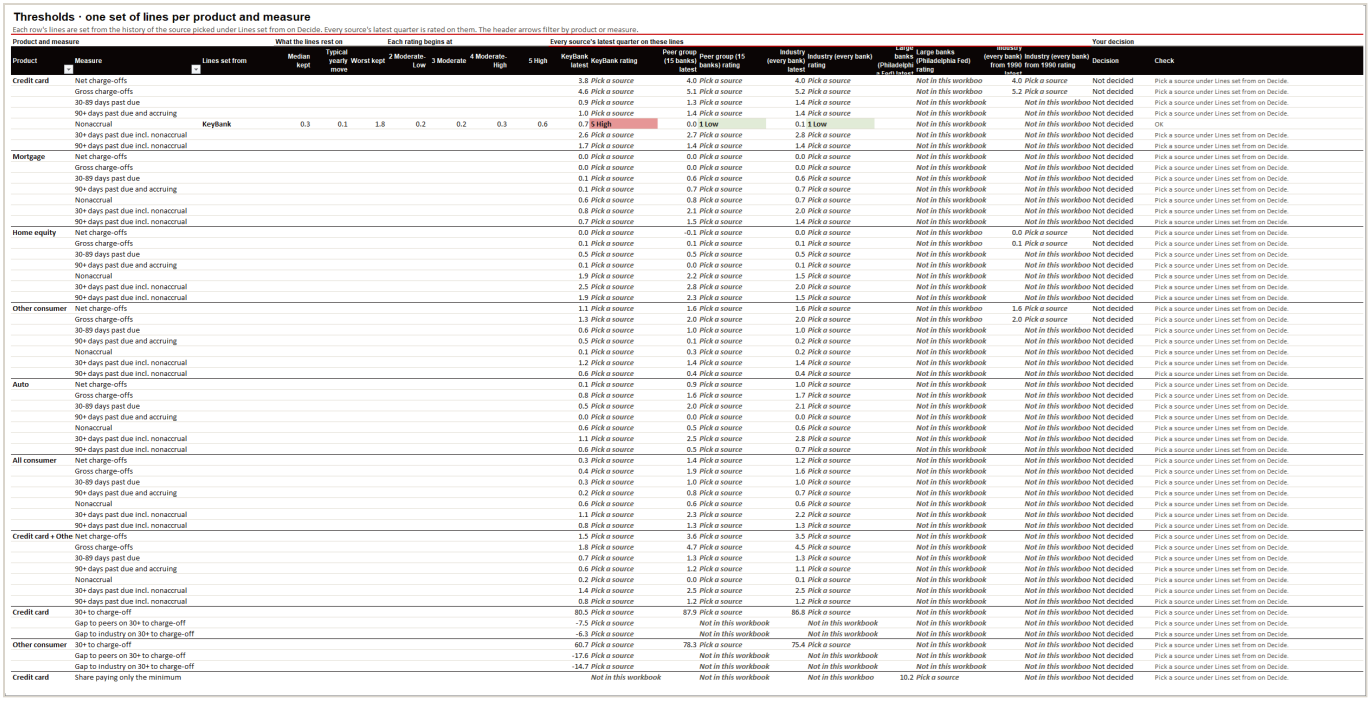
Width of Moderate	More quarters rate 3 Moderate.
Moderate-Low begins	Fewer quarters rate 1 Low.
How rare High is (z)	Fewer quarters rate 5 High. A z of 2.5 is rare; 3.5 is the usual line for an outlier.
On a line, rate it	With worse, a value exactly on a line takes the worse rating.

What fits the history the lines are set from

Width of Moderate	Fits below 1.00, where Moderate-Low begins. It fits.
Moderate-Low begins	Fits above 0.50, and below 2.98 so line 2 stays above zero. It fits.
How rare High is (z)	Fits above 0.34 (High beyond Moderate-High) and up to 10.98 (some quarter kept can rate High). It fits.
On a line, rate it	Either fits.

The solid line is the source the lines are set from. A dashed line is another source switched on under On chart, in the color of its name. A gray stretch is a period left out of the history the lines are set from: it is drawn, and it does not set the lines.
The settings apply to every source's history: the lines picked, and what each source's own history would give. The shading runs from 1 Low to 5 High. The chart starts at the first quarter any line has, within the window set under Chart shows. A diamond along the top marks a quarter in which a reporting change took effect (R

B3 · On Decide, click Credit card and Nonaccrual, and pick KeyBank under Lines set from (Decide tab, photographed by Excel during the walk)

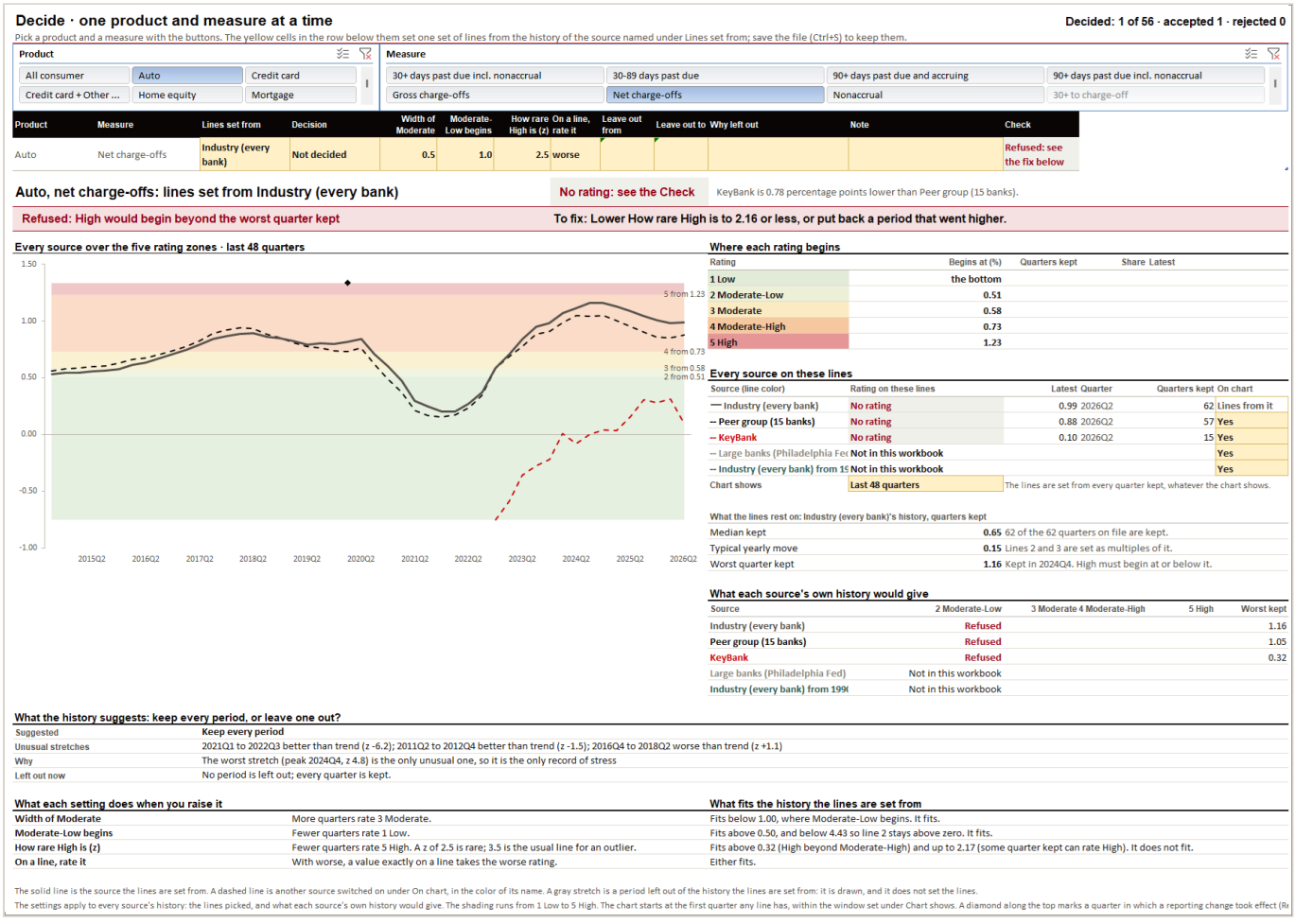


Thresholds · one set of lines per product and measure

Each row's lines are set from the history of the source picked under Lines set from on Decide. Every source's latest quarter is rated on them. The header arrows filter by product or measure.

Product and measure			What the lines rest on			Each rating begins at				Every source's latest quarter on these lines													
Product	Measure	Lines set from	Median kept	Typical yearly move	Worst kept	2 Moderate-Low	3 Moderate	4 Moderate-High	5 High	KeyBank latest	KeyBank rating	Peer group (15 banks) latest	Peer group (15 banks) rating	Industry (every bank) latest	Industry (every bank) rating	Large banks (Philadelphia Fed) latest	Large banks (Philadelphia Fed) rating	Industry (every bank) from 1990 latest	Industry (every bank) from 1990 rating	Decision	Check
Credit card	Net charge-offs									3.8	Pick a source	4.0	Pick a source	4.0	Pick a source		Not in this workboo	4.0	Pick a source	Not decided	Pick a source under Lines set from on Decide.
	Gross charge-offs									4.6	Pick a source	5.1	Pick a source	5.2	Pick a source		Not in this workboo	5.2	Pick a source	Not decided	Pick a source under Lines set from on Decide.
	30-89 days past due									0.9	Pick a source	1.3	Pick a source	1.4	Pick a source		Not in this workbook		Not in this workboo	Not decided	Pick a source under Lines set from on Decide.
	90+ days past due and accruing									1.0	Pick a source	1.4	Pick a source	1.4	Pick a source		Not in this workbook		Not in this workboo	Not decided	Pick a source under Lines set from on Decide.
	Nonaccrual	KeyBank	0.3	0.1	1.8	0.2	0.2	0.3	0.6	0.7	5 High	0.0	1 Low	0.1	1 Low		Not in this workbook		Not in this workboo	Not decided	OK
	30+ days past due incl. nonaccrual									2.6	Pick a source	2.7	Pick a source	2.8	Pick a source		Not in this workbook		Not in this workboo	Not decided	Pick a source under Lines set from on Decide.
	90+ days past due incl. nonaccrual									1.7	Pick a source	1.4	Pick a source	1.4	Pick a source		Not in this workbook		Not in this workboo	Not decided	Pick a source under Lines set from on Decide.
Mortgage	Net charge-offs									0.0	Pick a source	0.0	Pick a source	0.0	Pick a source		Not in this workbook		Not in this workboo	Not decided	Pick a source under Lines set from on Decide.
	Gross charge-offs									0.0	Pick a source	0.0	Pick a source	0.0	Pick a source		Not in this workbook		Not in this workboo	Not decided	Pick a source under Lines set from on Decide.
	30-89 days past due									0.1	Pick a source	0.6	Pick a source	0.6	Pick a source		Not in this workbook		Not in this workboo	Not decided	Pick a source under Lines set from on Decide.
	90+ days past due and accruing									0.1	Pick a source	0.7	Pick a source	0.7	Pick a source		Not in this workbook		Not in this workboo	Not decided	Pick a source under Lines set from on Decide.
	Nonaccrual									0.6	Pick a source	0.8	Pick a source	0.7	Pick a source		Not in this workbook		Not in this workboo	Not decided	Pick a source under Lines set from on Decide.
	30+ days past due incl. nonaccrual									0.8	Pick a source	2.1	Pick a source	2.0	Pick a source		Not in this workbook		Not in this workboo	Not decided	Pick a source under Lines set from on Decide.
	90+ days past due incl. nonaccrual									0.7	Pick a source	1.5	Pick a source	1.4	Pick a source		Not in this workbook		Not in this workboo	Not decided	Pick a source under Lines set from on Decide.
Home equity	Net charge-offs									0.0	Pick a source	-0.1	Pick a source	0.0	Pick a source		Not in this workboo	0.0	Pick a source	Not decided	Pick a source under Lines set from on Decide.
	Gross charge-offs									0.1	Pick a source	0.1	Pick a source	0.1	Pick a source		Not in this workboo	0.1	Pick a source	Not decided	Pick a source under Lines set from on Decide.
	30-89 days past due									0.5	Pick a source	0.5	Pick a source	0.5	Pick a source		Not in this workbook		Not in this workboo	Not decided	Pick a source under Lines set from on Decide.
	90+ days past due and accruing									0.1	Pick a source	0.0	Pick a source	0.1	Pick a source		Not in this workbook		Not in this workboo	Not decided	Pick a source under Lines set from on Decide.
	Nonaccrual									1.9	Pick a source	2.2	Pick a source	1.5	Pick a source		Not in this workbook		Not in this workboo	Not decided	Pick a source under Lines set from on Decide.
	30+ days past due incl. nonaccrual									2.5	Pick a source	2.8	Pick a source	2.0	Pick a source		Not in this workbook		Not in this workboo	Not decided	Pick a source under Lines set from on Decide.
	90+ days past due incl. nonaccrual									1.9	Pick a source	2.3	Pick a source	1.5	Pick a source		Not in this workbook		Not in this workboo	Not decided	Pick a source under Lines set from on Decide.
Other consumer	Net charge-offs									1.1	Pick a source	1.6	Pick a source	1.6	Pick a source		Not in this workboo	1.6	Pick a source	Not decided	Pick a source under Lines set from on Decide.
	Gross charge-offs									1.3	Pick a source	2.0	Pick a source	2.0	Pick a source		Not in this workboo	2.0	Pick a source	Not decided	Pick a source under Lines set from on Decide.
	30-89 days past due									0.6	Pick a source	1.0	Pick a source	1.0	Pick a source		Not in this workbook		Not in this workboo	Not decided	Pick a source under Lines set from on Decide.
	90+ days past due and accruing									0.5	Pick a source	0.1	Pick a source	0.2	Pick a source		Not in this workbook		Not in this workboo	Not decided	Pick a source under Lines set from on Decide.
	Nonaccrual									0.1	Pick a source	0.3	Pick a source	0.2	Pick a source		Not in this workbook		Not in this workboo	Not decided	Pick a source under Lines set from on Decide.
	30+ days past due incl. nonaccrual									1.2	Pick a source	1.4	Pick a source	1.4	Pick a source		Not in this workbook		Not in this workboo	Not decided	Pick a source under Lines set from on Decide.
	90+ days past due incl. nonaccrual									0.6	Pick a source	0.4	Pick a source	0.4	Pick a source		Not in this workbook		Not in this workboo	Not decided	Pick a source under Lines set from on Decide.
Auto	Net charge-offs									0.1	Pick a source	0.9	Pick a source	1.0	Pick a source		Not in this workbook		Not in this workboo	Not decided	Pick a source under Lines set from on Decide.
	Gross charge-offs									0.8	Pick a source	1.6	Pick a source	1.7	Pick a source		Not in this workbook		Not in this workboo	Not decided	Pick a source under Lines set from on Decide.
	30-89 days past due									0.5	Pick a source	2.0	Pick a source	2.1	Pick a source		Not in this workbook		Not in this workboo	Not decided	Pick a source under Lines set from on Decide.
	90+ days past due and accruing									0.0	Pick a source	0.0	Pick a source	0.0	Pick a source		Not in this workbook		Not in this workboo	Not decided	Pick a source under Lines set from on Decide.
	Nonaccrual									0.5	Pick a source	0.1	Pick a source	0.6	Pick a source		Not in this workbook		Not in this workboo	Not decided	Pick a source under Lines set from on Decide.
	30+ days past due incl. nonaccrual									1.1	Pick a source	2.5	Pick a source	2.8	Pick a source		Not in this workbook		Not in this workboo	Not decided	Pick a source under Lines set from on Decide.
	90+ days past due incl. nonaccrual									0.6	Pick a source	0.5	Pick a source	0.7	Pick a source		Not in this workbook		Not in this workboo	Not decided	Pick a source under Lines set from on Decide.
All consumer	Net charge-offs									0.3	Pick a source	1.4	Pick a source	1.2	Pick a source		Not in this workbook		Not in this workboo	Not decided	Pick a source under Lines set from on Decide.
	Gross charge-offs									0.4	Pick a source	1.9	Pick a source	1.6	Pick a source		Not in this workbook		Not in this workboo	Not decided	Pick a source under Lines set from on Decide.
	30-89 days past due									0.3	Pick a source	1.0	Pick a source	1.0	Pick a source		Not in this workbook		Not in this workboo	Not decided	Pick a source under Lines set from on Decide.
	90+ days past due and accruing									0.2	Pick a source	0.8	Pick a source	0.7	Pick a source		Not in this workbook		Not in this workboo	Not decided	Pick a source under Lines set from on Decide.
	Nonaccrual									0.6	Pick a source	0.6	Pick a source	0.6	Pick a source		Not in this workbook		Not in this workboo	Not decided	Pick a source under Lines set from on Decide.
	30+ days past due incl. nonaccrual									1.1	Pick a source	2.3	Pick a source	2.2	Pick a source		Not in this workbook		Not in this workboo	Not decided	Pick a source under Lines set from on Decide.
	90+ days past due incl. nonaccrual									0.8	Pick a source	1.3	Pick a source	1.3	Pick a source		Not in this workbook		Not in this workboo	Not decided	Pick a source under Lines set from on Decide.
Credit card + Othe	Net charge-offs									1.5	Pick a source	3.6	Pick a source	3.5	Pick a source		Not in this workbook		Not in this workboo	Not decided	Pick a source under Lines set from on Decide.
	Gross charge-offs									1.8	Pick a source	4.7	Pick a source	4.5	Pick a source		Not in this workbook		Not in this workboo	Not decided	Pick a source under Lines set from on Decide.
	30-89 days past due									0.7	Pick a source	1.3	Pick a source	1.3	Pick a source		Not in this workbook		Not in this workboo	Not decided	Pick a source under Lines set from on Decide.
	90+ days past due and accruing									0.6	Pick a source	1.2	Pick a source	1.1	Pick a source		Not in this workbook		Not in this workboo	Not decided	Pick a source under Lines set from on Decide.
	Nonaccrual									0.2	Pick a source	0.0	Pick a source	0.1	Pick a source		Not in this workbook		Not in this workboo	Not decided	Pick a source under Lines set from on Decide.
	30+ days past due incl. nonaccrual									1.4	Pick a source	2.5	Pick a source	2.5	Pick a source		Not in this workbook		Not in this workboo	Not decided	Pick a source under Lines set from on Decide.
	90+ days past due incl. nonaccrual									0.8	Pick a source	1.2	Pick a source	1.2	Pick a source		Not in this workbook		Not in this workboo	Not decided	Pick a source under Lines set from on Decide.
Credit card	30+ to charge-off									80.5	Pick a source	87.9	Pick a source	86.8	Pick a source		Not in this workbook		Not in this workboo	Not decided	Pick a source under Lines set from on Decide.
	Gap to peers on 30+ to charge-off									-7.5	Pick a source	Not in this workbook		Not in this workbook			Not in this workbook		Not in this workboo	Not decided	Pick a source under Lines set from on Decide.
	Gap to industry on 30+ to charge-off									-6.3	Pick a source	Not in this workbook		Not in this workbook			Not in this workbook		Not in this workboo	Not decided	Pick a source under Lines set from on Decide.
Other consumer	30+ to charge-off									60.7	Pick a source	78.3	Pick a source	75.4	Pick a source		Not in this workbook		Not in this workboo	Not decided	Pick a source under Lines set from on Decide.
	Gap to peers on 30+ to charge-off									-17.6	Pick a source	Not in this workbook		Not in this workbook			Not in this workbook		Not in this workboo	Not decided	Pick a source under Lines set from on Decide.
	Gap to industry on 30+ to charge-off									-14.7	Pick a source	Not in this workbook		Not in this workbook			Not in this workbook		Not in this workboo	Not decided	Pick a source under Lines set from on Decide.
Credit card	Share paying only the minimum									Not in this workbook		Not in this workbook		Not in this workboo		10.2	Pick a source		Not in this workboo	Not decided	Pick a source under Lines set from on Decide.

B3 · On Decide, click Credit card and Nonaccrual, and pick KeyBank under Lines set from (Thresholds tab, photographed by Excel during the walk)



Decide · one product and measure at a time

Decided: 1 of 56 · accepted 1 · rejected 0

Pick a product and a measure with the buttons. The yellow cells in the row below them set one set of lines from the history of the source named under Lines set from; save the file (Ctrl+S) to keep them.

Product: All consumer | Auto | Credit card | Credit card + Other ... | Home equity | Mortgage

Measure: 30+ days past due incl. nonaccrual | 30-89 days past due | 90+ days past due and accruing | 90+ days past due incl. nonaccrual | Gross charge-offs | Net charge-offs | Nonaccrual | 30+ to charge-off

Product	Measure	Lines set from	Decision	Width of Moderate	Moderate-Low begins	How rare High is (z)	On a line, rate it	Leave out from	Leave out to	Why left out	Note	Check
Auto	Net charge-offs	Industry (every bank)	Not decided	0.5	1.0	2.5	worse					Refused: see the fix below

Auto, net charge-offs: lines set from Industry (every bank)

No rating: see the Check — KeyBank is 0.78 percentage points lower than Peer group (15 banks).

Refused: High would begin beyond the worst quarter kept — To fix: Lower How rare High is to 2.16 or less, or put back a period that went higher.

Every source over the five rating zones · last 48 quarters

Where each rating begins

Rating	Begins at (%)	Quarters kept	Share	Latest
1 Low	the bottom			
2 Moderate-Low	0.51			
3 Moderate	0.58			
4 Moderate-High	0.73			
5 High	1.23			

Every source on these lines

Source (line color)	Rating on these lines	Latest	Quarter	Quarters kept	On chart
— Industry (every bank)	No rating	0.99	2026Q2	62	Lines from it
-- Peer group (15 banks)	No rating	0.88	2026Q2	57	Yes
-- KeyBank	No rating	0.10	2026Q2	15	Yes
-- Large banks (Philadelphia Fed	Not in this workbook				Yes
-- Industry (every bank) from 19	Not in this workbook				Yes
Chart shows	Last 48 quarters	The lines are set from every quarter kept, whatever the chart shows.			

What the lines rest on: Industry (every bank)'s history, quarters kept

Median kept	0.65	62 of the 62 quarters on file are kept.
Typical yearly move	0.15	Lines 2 and 3 are set as multiples of it.
Worst quarter kept	1.16	Kept in 2024Q4. High must begin at or below it.

What each source's own history would give

Source	2 Moderate-Low	3 Moderate	4 Moderate-High	5 High	Worst kept
Industry (every bank)	Refused				1.16
Peer group (15 banks)	Refused				1.05
KeyBank	Refused				0.32
Large banks (Philadelphia Fed)	Not in this workbook				
Industry (every bank) from 199(	Not in this workbook				

What the history suggests: keep every period, or leave one out?

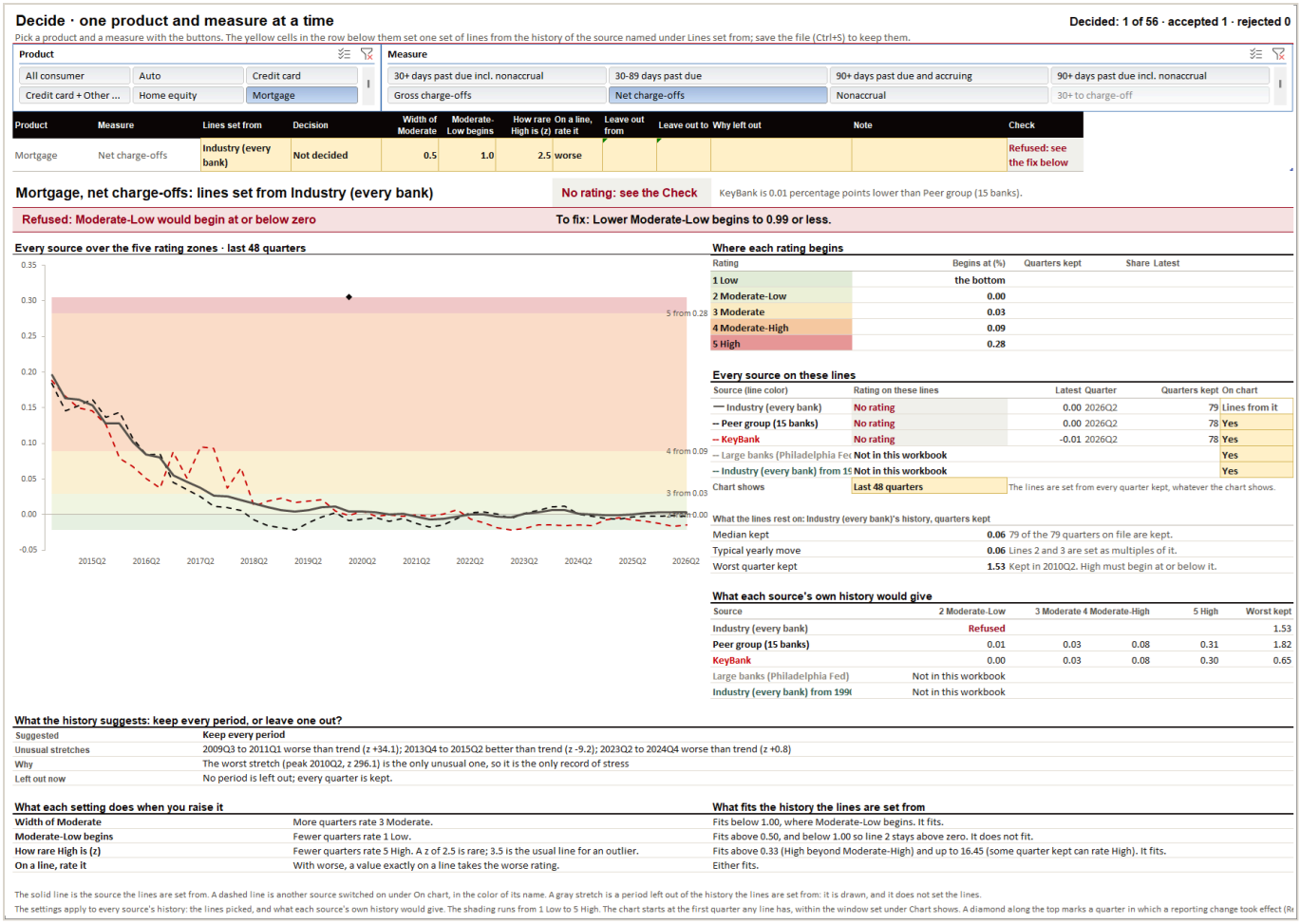
Suggested	Keep every period
Unusual stretches	2021Q1 to 2022Q3 better than trend (z -6.2); 2011Q2 to 2012Q4 better than trend (z -1.5); 2016Q4 to 2018Q2 worse than trend (z +1.1)
Why	The worst stretch (peak 2024Q4, z 4.8) is the only unusual one, so it is the only record of stress
Left out now	No period is left out; every quarter is kept.

What each setting does when you raise it		What fits the history the lines are set from
Width of Moderate	More quarters rate 3 Moderate.	Fits below 1.00, where Moderate-Low begins. It fits.
Moderate-Low begins	Fewer quarters rate 1 Low.	Fits above 0.50, and below 4.43 so line 2 stays above zero. It fits.
How rare High is (z)	Fewer quarters rate 5 High. A z of 2.5 is rare; 3.5 is the usual line for an outlier.	Fits above 0.32 (High beyond Moderate-High) and up to 2.17 (some quarter kept can rate High). It does not fit.
On a line, rate it	With worse, a value exactly on a line takes the worse rating.	Either fits.

The solid line is the source the lines are set from. A dashed line is another source switched on under On chart, in the color of its name. A gray stretch is a period left out of the history the lines are set from: it is drawn, and it does not set the lines.

The settings apply to every source's history: the lines picked, and what each source's own history would give. The shading runs from 1 Low to 5 High. The chart starts at the first quarter any line has, within the window set under Chart shows. A diamond along the top marks a quarter in which a reporting change took effect (R

B8 · Click Auto and Net charge-offs, and pick Industry (every bank) under Lines set from (Decide tab, photographed by Excel during the walk)



B9 · Click Mortgage and Net charge-offs, and pick Industry (every bank) (Decide tab, photographed by Excel during the walk)